NASA SPACE APPS CHALLENGE — TECHNICAL REPORT

Member 5: Physics-Based Water Balance Engine

Agro-meteorological Simulation, FAO-56 Evapotranspiration, and Root-Zone Irrigation Scheduling

1. Executive Summary & Objective Alignment

Requested Requirement	Target Specification	Implemented Solution & Result
Module Philosophy	Tested physics-based calculation, *not* an unsupported ML model	Deterministic FAO-56 Penman-Monteith physics framework; Zero black-box bias.
Atmospheric Water Demand	Reference Evapotranspiration (ET_0) & Crop Demand (ET_c)	Full energy-balance calculation taking 6 meteorological parameters from NASA POWER.
Hydraulic Dynamics	Root-zone depletion, Water Stress & Irrigation Triggering	Daily continuous soil balance modeling TAW, RAW, P_{eff}, and immediate replenishment.
Input Integration	Member 2 (Weather) + Member 4 (Crop K_c)	Seamlessly integrates 5-year NASA POWER series (2019–2024) with crop-specific coefficients.

2. Governing Physics & Mathematical Formulations

A. FAO-56 Penman-Monteith Equation (Reference Evapotranspiration - ET_0)

$$ET_0 = [\ 0.408\Delta(R_n - G) + \gamma\ (900\ /\ (T_{mean} + 273))\ u_2\ (e_s - e_a)\ ]\ /\ [\ \Delta + \gamma\ (1 + 0.34\ u_2)\ ]$$

Where Δ is saturation vapor pressure slope, R_n is net radiation, γ is psychrometric constant, u_2 is wind speed at 2m, and $(e_s - e_a)$ represents vapor pressure deficit. Daily ground heat flux $G \approx 0$.

B. Crop Evapotranspiration & Available Water Constants

$$ET_c = K_c \times ET_0 \quad | \quad TAW = 1000 \times (\theta_{FC} - \theta_{WP}) \times Z_r \quad | \quad RAW = p \times TAW$$

TAW = Total Available Water (mm), RAW = Readily Available Water (mm), θ_{FC} = Field Capacity (0.30), θ_{WP} = Wilting Point (0.15), Z_r = Root Depth (0.6m), p = Depletion Fraction (0.50).

C. Daily Root-Zone Water Balance & Irrigation Trigger

$$D_{r,\ t} = D_{r,\ t-1} - P_{eff,\ t} + ET_{c,\ t} \quad [\ \text{If } D_{r,\ t} > RAW \rightarrow \text{Trigger Irrigation: } I_t = D_{r,\ t}\ ]$$

$D_{r,t}$ is root-zone depletion at day t. Effective precipitation P_{eff} is calculated at 80% efficiency for rain > 2mm. When depletion exceeds RAW, water stress occurs and irrigation resets D_r to 0.

3. Execution Algorithm Pipeline

1. **Data Ingestion:** Ingest daily time-series from NASA POWER API (T_{max}, T_{min}, RH, Wind Speed, Solar Radiation, Precipitation).
2. **Psychrometric Conversion:** Compute atmospheric pressure (P_{atm}), psychrometric constant (γ), saturated vapor pressure (e_s), and actual vapor pressure (e_a).
3. **Radiation Balance:** Estimate net shortwave (R_{ns}) via albedo ($\alpha = 0.23$) and net longwave (R_{nl}) via Stefan-Boltzmann law and cloudiness fractions.
4. **Depletion Tracking:** Advance daily soil reservoir status ($D_{r,t}$). Flag waterlogging when intense precipitation saturates storage capacity.
5. **Stress Detection & Advisory:** Detect when $D_{r,t}$ > RAW, issue `DROUGHT_WATER_STRESS_IRRIGATE` alert, and compute exact millimeter irrigation depth.

4. Final Deliverables Summary

Delivered Artifacts:

- `water_balance_simulation.csv`: 1,827 days (5 full years) of verified daily ET_0, ET_c, depletion, and irrigation event logs.
- `fao56_water_balance_curve.png`: High-resolution hydrological trajectory showing depletion peaks and threshold resets.
- `member_5_physics_water_balance_code.zip`: Standalone modular Python engine, Colab notebook, and technical API documentation.